

- Q.24 What are the advantages of electrical energy over other types of energy?
- Q.25 Explain the concept of resistance and inductance?
- Q.26 Explain isolation transformer.
- Q.27 Explain the application of transformers with examples.
- Q.28 Explain submersible and flame proof motors and their applications.
- Q.29 How the identification of wiring system is done?
- Q.30 Explain the concept of fuses and their application.
- Q.31 What are common safety measures in domestic installation?
- Q.32 What is a thyristor and its uses?
- Q.33 How the load of batteries is maintained?
- Q.34 How AC and DC motors are identified?
- Q.35 Describe voltage and current relationship in AC and DC supply.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the Lenz Rule and Fleming's rules. Explain power factor and method to improve it?
- Q.37 With the help of a diagram describe low voltage and high voltage transmission.
- Q.38 Explain how identification of three phase wires is done. Draw domestic supply diagram.

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3rd Sem / Aircraft Maintenance Engg. Sub : Elements of Electrical & Electronics Engineering-I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which instrument is used to measure electrical power in a circuit?
a) Ammeter b) Voltmeter
c) Wattmeter d) Frequency Meter
- Q.2 What is the function of a frequency meter?
a) To measure the voltage of an electrical circuit
b) To measure the current flowing through a circuit
c) To measure the frequency of an electrical signal
d) To measure the power factor of an electrical load
- Q.3 Which part of a DC generator is responsible for converting mechanical energy into electrical energy?
a) Field winding b) Armature winding
c) Commutator d) Brushes
- Q.4 Which component of an AC generator is responsible for converting mechanical energy into electrical energy?
a) Stator b) Rotor
c) Armature d) Commutator
- Q.5 What is the primary difference between a single-phase and multi-phase AC generator?
a) Single phase generators have a single coil in the

stator, while multi-phase generators have multiple coils.

- b) Single-phase generators produce single alternating current, while multi-phase generator produce multiple alternating currents.
- c) Single-phase generators are used for residential applications, while multi-phase generators are used for industrial applications.
- d) Single phase generators have a higher voltage output compared to multi-phase generators

Q.6 What is the primary function of battery testing?

- a) To determine the voltage rating
- b) To assess the battery's physical construction
- c) To evaluate the battery's capacity and performance
- d) To measure the battery's temperature

Q.7 What is the primary function of a band pass filter?

- a) To allow a specific band of frequencies to pass through
- b) To amplify all frequencies equally
- c) To block all frequencies except a specific band
- d) To attenuate all frequencies

Q.8 In an A.C. generator, increase in number of turns in the coil_____.

- a) Increases emf
- b) Decrease emf
- c) Makes the emf Zero
- d) Maintains the emf at a constant value

Q.9 While charging, what happens to the current in the capacitive circuit?

- a) Decreases linearly

- b) Increases linearly
- c) Decrease exponentially
- d) Increases exponentially

Q.10 In a sinusoidal wave, average current is always _____ rms current.

- a) Greater than
- b) Less than
- c) Equal to
- d) Not related

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What is the motor used for driving pump?

Q.12 What are maintenance free batteries?

Q.13 What is the full form of CVT.

Q.14 What is alternating EMF?

Q.15 What is the Primary difference between AC and DC?

Q.16 What is peak factor?

Q.17 What are some of the losses in transformer?

Q.18 What are the applications of servo motor in process control?

Q.19 What is the role of Isolation Transformer?

Q.20 What does Auto transformer do?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What is power factor and the role of capacitor in power factor?

Q.22 Explain the concept of star and delta connection.

Q.23 How the various quantities of electricity are measured?